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Bovine facial recognition through muzzles

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Abstract

This review examines the application of neural network architectures for cattle identification and recognition using snout images as the primary biometric trait, motivated by the anatomical uniqueness and high inter-individual variability of this region. Beyond reporting performance metrics, the analyzed literature consistently highlights that the discriminative power of bovine facial recognition systems is strongly associated with specific visual features, including the spatial configuration of nostrils, texture patterns of the muzzle surface, local curvature of the snout region, and contrast variations under different illumination conditions. A wide range of algorithms and tools, such as SIFT, SURF, KNN, and deep learning architectures including YOLO-based detectors and ResNet-50 backbones, have been employed to extract and model these features effectively. The reviewed studies report high recognition performance, with recent YOLOv8-based approaches achieving accuracies close to or equal to 100% under controlled conditions. Furthermore, the use of transfer learning, convolutional feature hierarchies, and ensemble classifiers, such as random forests, enhances robustness to pose variation, noise, and environmental variability. These findings demonstrate that the success of bovine facial and muzzle recognition is primarily driven by the quality and stability of learned visual features rather than by dataset-specific configurations. As a result, such systems show strong potential for real-time livestock monitoring, health assessment, disease tracking, and scalable precision livestock management, with applicability extending to other animal species.

Keywords: *cattle recognition, precision livestock management, neural networks, facial recognition, muzzle recognition*

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1. Introduction

Machine learning methods are increasingly being used not only for prediction through static models but also as highly integrated components in more dynamic and real-time decision-making processes. In this context, advances in computer vision have enabled remote sensing across various sectors of society, ranging from biometrics used in facial recognition and security devices to the operation of household appliances and Internet of Things equipment, as well as energy supply management. An example of this is provided by [1], who employed photogrammetry for feature detection in power distribution line corridors within the electrical sector, using machine learning techniques to enhance efficiency and sustainability.

Another application of machine learning has been marked by its economic impact on agriculture. The use of deep learning algorithms has proven effective for the rapid and accurate identification and monitoring of crops, based on image information and similarity analysis, as a strategy to mitigate crop damage and the spread of diseases [2].

In other areas of the agricultural sector, there are also consistent contributions, such as in aquaculture, which supports global fish production by eliminating bottlenecks along the production chain. This includes tasks like the counting and selection of fry, which traditionally depend exclusively on human intervention but become increasingly complex in dense and highly overlapping populations. Machine learning and artificial intelligence have emerged as technological tools with high problem-solving capacity to address challenges across different domains of captive species cultivation [3].

In response to the demands of the domestic market and export requirements aimed at ensuring food security, Brazil has been investing in technology to strengthen precision agriculture and livestock farming. The latter, focused on the meat production chain, has incorporated machine learning applications as a means to ensure the health and welfare of cattle herds. The use of algorithms and mobile web applications (apps) has proven to be a fast and efficient approach for data collection and recognition, enabling the modeling and evaluation of animals for disease prediction in beef cattle production [4].

The traceability enabled by the integration of machine learning and artificial intelligence is also an important component in cattle confinement systems, as it allows the individual identification of animals. The collected information focuses on sanitary control and the tracking of diseases. In contrast to conventional traceability methods such as electronic ear tags, microchips, buttons,

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or other noninvasive accessories like sensors and GPS collars, which remain artificial attachments, animal biometrics operate by identifying unique muzzle patterns comparable to human fingerprints. This method is entirely noninvasive and depends exclusively on machine learning modeling [5].

Identification through the traceability of bovine muzzle patterns has already emerged as a factor contributing to animal quality and welfare. However, barriers still exist to the implementation of a practical precision monitoring system on a large scale for extensive herds, encouraging research aimed at developing autonomous biometric identification approaches for cattle [6].

Undoubtedly, it is a fundamental element in precision livestock farming, and image capture has established itself as a useful and reliable approach for monitoring and automatically analyzing animal behavior, thereby enabling efficient management. Although pixel-wise image labeling may appear simple, it is a time-consuming task due to the irregular contours of each animal and the variations in posture during image acquisition, making segmentation a basic prerequisite for cattle identification. In this context, deep learning enables the labeling of a large number of images for recognition and, through single-image learning, the segmentation of each image [7].

Studies conducted by Awad [8] demonstrated the importance of traceability in cattle and buffalo as a means of identification, disease tracking, vaccination control, and production management. They also highlighted that identification methods based on ear tags, brands, tattoos, and electrical techniques are susceptible to fraud, duplication, and security challenges. Among image-based biometric identifiers, muzzle prints, iris patterns, and retinal vascular patterns are mentioned. However, when determining the most suitable approach, it is necessary to consider key challenges for computerized systems, such as identification accuracy, processing time, and operability.

Machine learning has shown considerable potential for biometric recognition in livestock, producing high-quality data that can enhance herd management and monitoring. However, for these models to be practically useful, they must be easily deployable in real-world farm environments and provide a comprehensive understanding of the underlying patterns they aim to detect. Efficient model deployment allows for the optimal use of computational resources, facilitating innovation in livestock management and supporting the sustainable development of modern animal husbandry. The integration of such technology can improve decision-making, reduce manual effort, and enable more precise interventions to maintain herd health and productivity.

In order to assess the effectiveness of neural network architectures for the identification of bovine muzzle patterns, this study presents a systematic review of the literature from the past five years. The review aims to identify which approaches to cattle recognition through muzzle biometrics achieve the highest accuracy and precision while minimizing errors caused by phenotypic similarities, such as resemblances in observable characteristics of color, shape, and size. By synthesizing these findings, the study provides insights into the most reliable methods for herd monitoring and management, highlighting the practical implications of implementing biometric technologies in livestock operations and offering guidance for future research in this domain.

2. Materials and methods

The objective of this study is to conduct a systematic literature review covering the past five years, from 2019 to 2023. Following data collection, a quantitative analysis was performed to assess the volume of scientific publications and to examine the elements presented, thereby contributing to the scientific knowledge within the specific fields investigated. Another notable aspect concerns the indicators analyzed in the post-review studies and their contribution to aligning public policies with animal and human health and safety measures.

The investigation began with the construction of a descriptive theoretical framework on the use of machine learning and its relationship with veterinary science and livestock farming. The results were collected from scientific studies indexed in electronic databases and were subsequently grouped according to their content.

Data collection was conducted between April and May 2024, with inclusion criteria based on updates of full scientific articles aligned with the objectives of this study and published in open access in the English language. Statistical metrics for document selection were constructed according to the descriptive terms used in the research databases available on ArXiv.org, DOAJ (Directory of Open Access Journals), Free EJournal, IEEE Electronic Library Journal, Journal@Ovid, Open Access, PubMed, PubMed Central, Science Direct, Social Science Citation Index (WOS), Springer Nature, Web of Science, Wiley Online Library, and World Scientific Journal, complemented by studies accessible through the Google Scholar platform.

The semantic connectors applied were used in the following combinations: muzzle and recognition, muzzle, muzzle recognition. This search yielded a total of 15,800 articles, of which, after applying the criteria of full open-access articles in English and based on the review of titles relevant to the research focus, 14,913 articles were excluded for not meeting the established criteria. This exclusion process involved the removal of duplicate articles, publications in foreign languages without an English translation, and those originally published in Asian languages, resulting in a total of 887 articles. This stage was designated as the first selection filter, in which a preliminary choice was made for subject categorization and collection of documents that met the inclusion criteria.

The second filter assessed the content of the topics addressed in the abstracts and the identification of each document according to categorization guided by semantic groups. **Figure 1** illustrates the number of abstracts selected for review according to the databases searched. Subsequently, duplicate publications or works considered supplementary and of minimal relevance to the research objective were discarded, resulting in a total of 26 articles. In a more detailed screening, an additional 17 articles were included; however, 4 did not present conclusive results, leaving a total of 39 documents. The collected materials were processed and analyzed using the open-source software IRaMuTeQ [9], which enabled the efficient and in-depth processing and analysis of large volumes of text, preserving the contextual relationships directed by the semantic connectors and avoiding mere mechanical descriptions of the studied content.

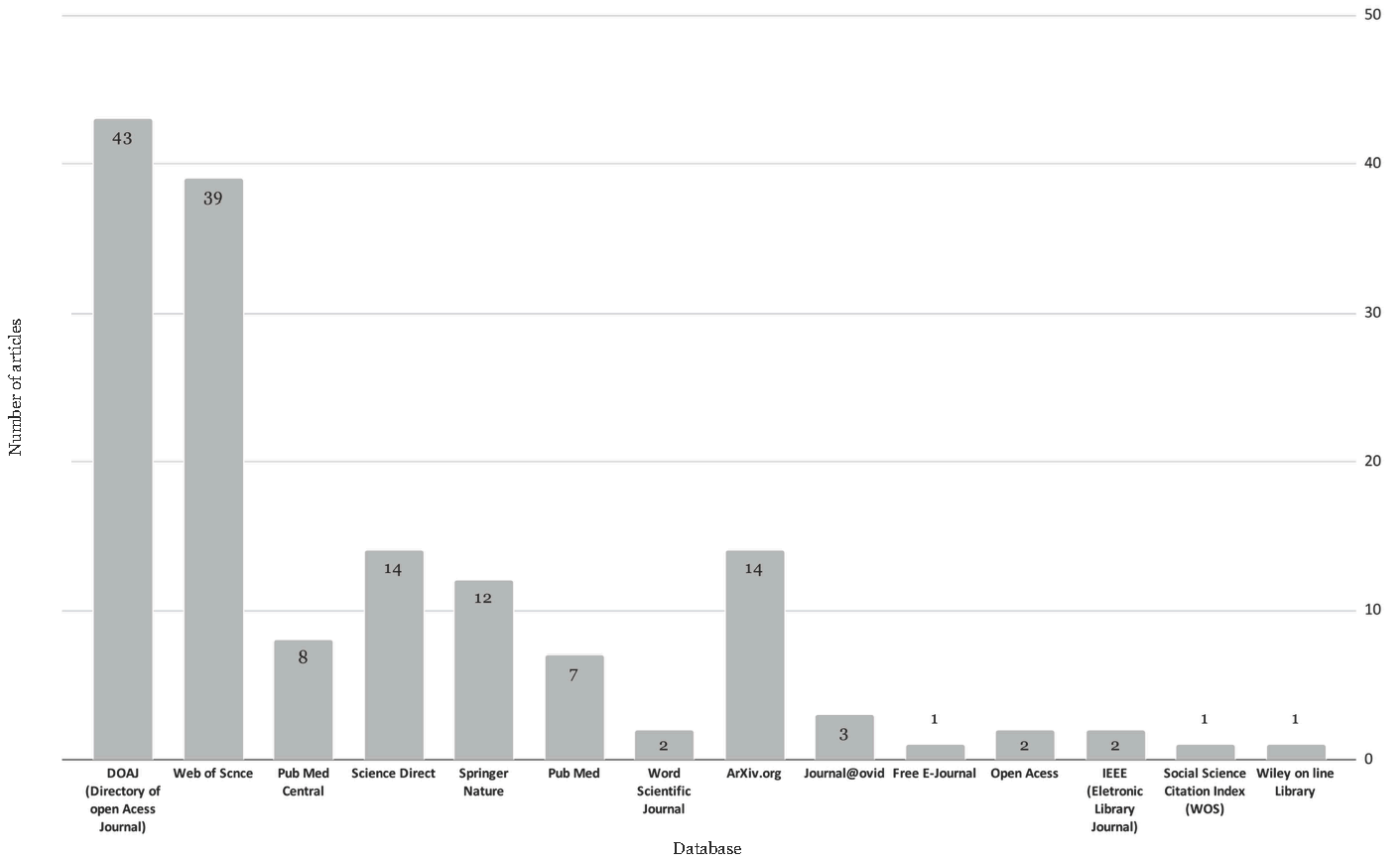


Figure 1 • Distribution of the selected articles in the preliminary research across databases and search platforms by binary combination from 2019 to 2023.

A basic lexicographical analysis was applied to determine word frequency calculations within the study's focus, including the total number of words, average frequency, and the number of hapax occurrences. This analysis aimed to characterize the vocabulary and establish a dictionary of reduced forms, identifying active and supplementary forms applied in each study, thereby enabling proper grouping.

The results of the statistical analysis of the 39 articles comprising the textual corpus revealed 412 occurrences of common words and 156 grammatical forms. The graphical representation of the lexicography indicated a higher frequency of terms within intervals above fifty, with one to two overlapping terms. Regarding hapax or terms occurring only once, a total of one hundred instances were observed, of which 24.7% were isolated occurrences and 64.10% were grammatical forms. These were primarily observed within frequency ranges of 10 to 20, as shown in **Figure 2**, with an average of 10.30 lexical occurrences among the terms.

Given the wide variety of terms, it was not possible to establish an adequate representation based on similarity. All cited terms, regardless of their frequency, are presented in alphabetical order in **Table 1**. Considering the terms "muzzle" and "recognition" as central to the classes of active, supplementary, or combined forms, the database comprised 156 grammatical forms, forming five distinct subject groups addressing architecture models, recognition approaches, types of technology, biometric applications, and cattle muzzle recognition.

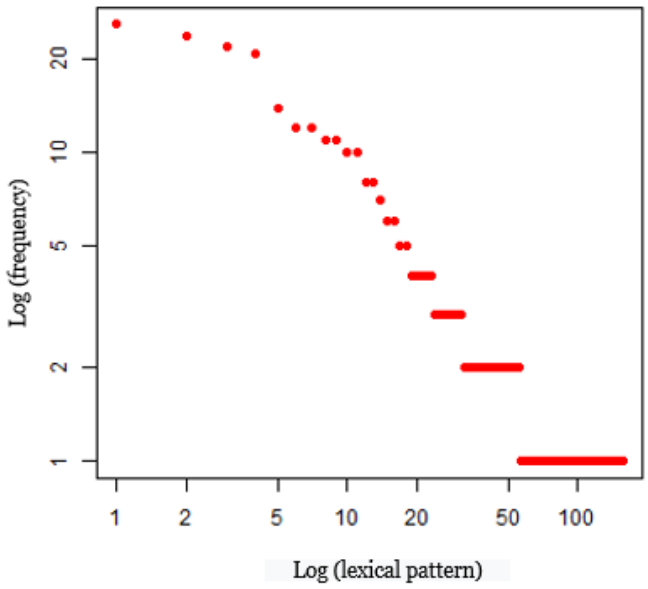


Figure 2 • Example of facial segmentation of the animals.

Table 1 • Description of the cited terms in alphabetical order.

accuracy	bovine	drive	horse	marker	precision	SNP
agriculture	brute	edge	humanity	matcher	print	sparse
algorithm	cation	efficient net	ICA	metric	process	SURF
animal	cattle	equine	identifique	mflops	provide	system
animality	classification	extraction	identification	microsatellite	radio	tag
architecture	CNN	face	identifier	MLP	random	task
artificial	cognitive	farm	image	model	RANSAC	time
autoencoder	compression	feature	individual	monitor	recognition	tomasi
automate	compute	field	inspire	muzzle	reduction	tracking
background	convolution	filter	intelligence	network	representation	transfer
bag	convolutional	flann	intelligent	neural	saen	tree
ballistics	cow	flow	intermediate	NN	schlieren	verification
barcodes	dairy	FLPP	kamphaeng	orient	science	vibrissa
base	database	force	LBP	owner	SDAE	vision
beef	DNB	forest	LDA	pantaneira	semiotic	visual
behavior	decision	frequency	learn	pattern	sense	visualization
bio	deep	gabor	livestock	PCA	sensor	VLAD
bioactive	depth	genotyping	LPP	perception	shi	welfare
biometric	detection	group	machine	personhood	shoot	word
blast	dog	hanwoo	management	phenotypic	sift	YOLO
body	drag	health	map	point	size	YOLO 3

3. Results and discussion

The selected articles (n = 39) were categorized following analysis for the construction of the literature review and grouped according to the chosen keywords and their concordance within the database. The relationship of these articles is detailed in **Table 1** and Figures S1 and S2 from the supplementary materials, which explains the categorization of the distinct groups and their distribution in detail. Continuing the lexical analysis and using the same word database from the similarity analysis, but focusing on the visual representation of the most frequently occurring isolated words within the textual corpus without considering their correlations, a word cloud was created, as shown in Figure S3 from the supplementary materials, also included in the supplementary materials.

Aligned with the proposed objective and aimed at deepening the understanding of the selected texts, adaptations were made following the guidelines presented by [10] on machine learning and techniques for cattle identification, resulting in the formulation of three guiding questions:

- Which techniques have been employed for animal identification over the past five years?
- Which Machine Learning and Deep Learning models have been applied in livestock farming to identify and recognize animals?
- Focusing on cattle farming, which methods and evaluation metrics are being used to identify and recognize livestock, regardless of their location?

Considering the distribution of isolated words and their citation frequency in the new selection linked to the guiding questions, the most prevalent terms in descending order were identification, cattle, muzzle, and learning, as shown in **Figure 3**. It is also important to note that no experimental data were generated, and all values were extracted from the consulted studies.

Although phenotypic aspects are commonly used as a reference for applying biometrics in animal identification, patterns such as coat markings often present gaps that the human eye cannot discern. To mitigate this challenge, computer vision techniques and image processing contribute significantly to capturing and removing artifacts, enabling the creation of a dataset that establishes biometric characteristics for each individual animal. The mathematical accuracy of the data is evaluated based on the statistics of each employed method, providing quality and reliability for precise analysis of the captured images [11].

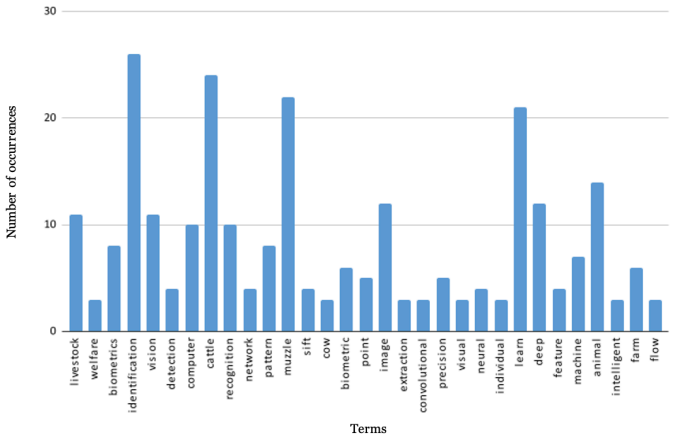


Figure 3 • Distribution of the most frequent single words.

The analytical discussion of the selected literature texts was grouped according to method, application, and accuracy for facial

and muzzle recognition of cattle using machine learning architectures and tools. Each study was further categorized into subgroups defined by the technique investigated, as shown in **Table 2**. However, for a more rigorous and structured discussion, each term and each acronym presented in the table, which correspond to the techniques employed in the reviewed studies, will be further clarified and cited accordingly in the subsequent discussion. The texts will first be presented by examining the foundational principles of each technique and its applications in animal identification, followed by a detailed explanation of their specific use in cattle identification as described in the respective works.

Table 2 • Methodological grouping of machine learning approaches applied to cattle face and muzzle recognition, organized according to the employed techniques and target applications.

Method	Purpose of use	References
CNNs	Identification of facial features and muzzle with SIFT	[12, 13]
DBN, SDAE, CLAHE	Identification with few images; Landmark predictor	[14, 15]
KNN, SIFT, SURF	Extraction of facial and muzzle features	[12, 16, 17]
ResNet + PointNet	Robustly identify specific muzzle patterns	[11, 18]
ResNet50	Discriminatory features of the face and muzzle	[19]
SDAE + DNB	Structure of the image pattern of the muzzle and face, contour and perception of beads and crest	[20]
SIFT	Noise or blurred image with data refinement	[21]
SURF, SIFT, Dense-SIFT + (χ2)	Multilayer identification and outlier removal	[12]
SURF; MSER	Recognition in a mixed set of animals	[22]
SVM	Large herds with lighting and multiscale variability	[6]
VGG16_BN; Shuffle NetV2	Muzzle printing in field herds	[23]
YOLO (v3) + SIFT; ResNetV2	Different image quality and consistency at angles	[24]
YOLO(v7)	Real-time capture of animals, varying lighting, nonlinear movement	[25]
YOLOv3 + ResNet50	Use of few images for individual identification of cattle	[5]
YOLOv5n + ConvBlock1	Interesting points and random sample	[26]
YOLOv8	Color depth, pimples and dermal projections	[27]

3.1. Study of the technique and animal identification

Visual biometrics refers to the identification of individuals based on distinctive visual characteristics and has gained increasing relevance in applications involving animal recognition and identification, particularly in the context of wildlife conservation and endangered species monitoring [28]. Within this framework, the concept of a "face" is understood as an identity-bearing representation grounded in phenotypic similarity. Etymologically, the term rostrum, derived from Latin, encompasses the frontal anatomical structures of the head that constitute the facial region in mammals, birds, and insects. The presence and morphological configuration of structures such as muzzles and beaks enable the categorization of animal groups and the discrimination of intra-group variability based on unique anatomical traits [29].

Illustrative examples include the pronounced facial differences observed between dog breeds such as bulldogs and German shepherds, as well as between Shih Tzus and Yorkshire Terriers. In veterinary and livestock management contexts, biometric marker systems support more efficient identification, monitoring, and traceability, contributing to vaccination control, disease management, and the mitigation of theft and fraud. Among the biometric traits commonly explored in animal identification studies are the iris, retina, muzzle, and body patterns [30].

It is worth noting that the more sensitive a biometric marker is, the greater the requirement for technologies that rely on more expensive equipment and systems, which directly correlates with cost. Such technologies demand higher-quality components and more advanced and rigorous processes to achieve precision and reliability. However, the adoption of cutting-edge or advanced technologies carries the risk of operational bottlenecks and the constant need for specialized expertise. Therefore, the use of solutions that employ reliable and precise markers through simpler and lower-cost approaches, which do not require highly specialized personnel for daily application and use, is highly advantageous.

Unique patterns and discriminative body markings used for individual identification are already well established for manta rays, penguins, whales, sharks, and the stripe patterns of tigers and zebras. Based on this knowledge, adaptations have been made for the biometric recognition of chimpanzees, lemurs, cattle, sheep, horses, and pigs [6]. Detection and recognition approaches are particularly important when monitoring species that inhabit hard-to-reach areas, and their effectiveness can be enhanced through the use of satellite imagery for detecting, identifying, and counting elusive species, whether of medium or large size.

Studies conducted by [31] explored biometric identification using technologies previously applied to human recognition, such as fingerprint, facial, and retinal pattern analysis, now adapted for animal identification. The muzzle has shown promising results as a primary biometric feature for recognizing mammals, including cattle, goats, sheep, and horses, providing not only a relevant ID for tracking but also additional information for disease control, monitoring of breeding and management programs, as well as enhancing animal welfare and sustainable production efficiency. For identification, object detection was performed using Yolo (v7),

achieving an average accuracy of 99.5% for face detection and 99.7% for muzzle detection. Feature filtering and extraction to mark points of interest employed SIFT (Scale-Invariant Feature Transform), followed by FLANN (Fast Library for Approximate Nearest Neighbors).

The system set demonstrated precise recognition capabilities and potential applications in animal security and health management. The process was carried out in three stages: detection of the animal and capture of the face in the image, localization of the animal, and detection of the muzzle or muzzle points within the facial image. Among the observed animal groups, horse biometrics achieved 100% accuracy in matching the muzzle point dataset, referred to as THDD (Horse Dataset). However, the presence of noise or blurring in the images due to animal movement was noted, necessitating feature filtering and extraction using SIFT, followed by FLANN for data refinement [19].

Advances in animal biometrics have enabled significant progress in livestock farming aimed at food supply, providing new perspectives for the management and promotion of animal and human health, as well as considerable socioeconomic importance. Following the guidelines of the WOAH (World Organization for Animal Health) regarding animal tracking systems, biometrics is considered essential and should replace traditional identification methods such as wool marks, tattoos, ear notches, and plastic or metal ear tags, which increase costs, particularly when applied to large herds of sheep, goats, pigs, and cattle [30].

Consequently, regarded as a pioneering milestone in computer vision, biometrics combined with general animal monitoring in livestock farming allows for pattern recognition and cognitive science applications as essential elements for the accurate recording, identification, and verification of animals [19].

However, due to the nature of cattle and pig herds and the lack of standardized comparison parameters, high-resolution images often allow only basic detection. Consequently, methods are designed based on computer vision and deep learning, facilitating the localization, tracking, and identification of animals through the direct application or combination of multiple approaches. Detection can be implemented using techniques such as Histograms of Oriented Gradients (HOGs), boosted cascades, and particle filters, alongside object tracking algorithms and Kalman filters. Among machine learning methods, Local Binary Patterns (LBPs) are used for classification, Fisher Linear Projection Preservation (FLPP) for facial and muzzle recognition, and Support Vector Machines (SVMs) for classification. Regarding architectures, R-CNN and YOLO are widely applied due to their speed, along with RetinaNet and detectors such as DetectoRS, SOLOv2, DETR, and Mask R-CNN for processing [32].

Feature extraction, being a process used to obtain a unique characteristic of the object in focus, allows for noise correction and the enhancement of results. Among the unique identifier patterns used in studies with animals having few images, several general-purpose extraction algorithms can be employed, such as SIFT, SURF, and BRISK (Binary Keypoints). For scalability, ORB (FAST-oriented and BRIEF-rotated) can also be applied [28].

Based on this premise, Xu et al. [14] proposed a method for the identification of cows using a small number of images as a measure to reduce the costs of herd recognition and localization for 243 animals. The approach employed a YOLOv5 detector augmented with a Ghost module and the MAML framework (Model-Agnostic Meta-Learning), enabling rapid adaptation to recognize individuals even with few images. An autoencoder was adopted to allow the Base-Learner to learn general features by combining supervised and unsupervised approaches. The proposed localization model achieved a mean Average Precision (mAP) of 99.5% and an identification accuracy of 90.43% with only five samples per animal.

Previous information from biometric analyses for the identification and recognition of cattle conducted by [15] indicates that analyzers such as ImageNET and COCO serve as benchmarks for deep learning models; however, the available facial datasets do not always provide specific elements capable of generating optimal mapping. The study recommends the application of trained models beginning with a face detector, followed by the use of a landmark predictor to establish reference points that act as anchors to rotate and align the image in a portrait position. This process assists in calculating the mean squared error (MSE) to measure the differences between the predicted coordinates and the original points, and finally to encode the image. YOLOv5 was identified as the base algorithm, enabling integration of the data within a score range of 0 to 1 and providing confidence in the available dataset.

Through modifications to the OSNet architecture, originally developed for human identification, accuracy was adapted for the problem of cattle recognition. Using a deep learning approach, the system can distinguish animals from images to identify lost livestock or prevent insurance fraud. The replacement of OSNet's convolutional layers with depthwise separable convolutions and rectified linear units (ReLUs), combined with a nonlinear activation function, enabled the addition of two attention modules while maintaining the same experimental conditions. This modification achieved higher accuracy compared to the original architecture, without compromising processing speed and while retaining compact storage capability [33].

3.2. Application of the technique in cattle farming

Historically, livestock management technologies have progressed from predominantly manual identification procedures to automated systems enabled by machine learning, allowing individual cattle recognition based on body contours, facial features, and muzzle patterns. Building upon architectural adaptations such as those applied to OSNet, recent studies have demonstrated that deep learning frameworks can be effectively tailored to cattle identification without compromising computational efficiency.

In dairy production systems, these recognition technologies further extend beyond identity verification, supporting the assessment of movement behavior, water and feed intake, rumination patterns, estrus detection, and the early identification of health conditions, including lameness and mastitis. Within this context, architectures and tools such as ArcFace (Additive Angular Margin Loss), RetinaFace-Mobilenet, and Cattle-FaceNet are frequently reported. Additionally, the application of convolutional neural networks, including Res52Net with task-specific adaptations, achieved recognition rates of 91.3% at a processing speed of 24 images per second, which increased to 94.53% through the integration of GPN features with the core ResNet50 architecture, illustrating the balance between accuracy gains and real-time applicability [12].

Continuing with the findings of [22], recognition accuracy values exceeded the benchmarks, reaching 95.9% and 97.4% when using three layers of the YOLOv3 algorithm. For muzzle print feature extraction, SIFT combined with RANSAC (Random Sample Consensus) was employed to remove outliers and enhance robustness. Local Binary Pattern (LBP) was also tested for extracting muzzle print features, achieving an accuracy of 93.3%, and when combined with classifiers such as SVM (Support Vector Machine) and KNN (K-Nearest Neighbor), particularly in the combination of KNN with SIFT and SURF, recognition accuracy reached 99.5%.

Comparing the reported recognition and accuracy percentages for convolutional networks Res52Net and ResNet50, and subsequently for YOLOv3 combined with the RANSAC extractor and classifiers including LBP, SVM, KNN, SIFT, and SURF in dairy cattle herds, these values were superior to those produced by Multilayer Perceptron (MLP), Random Forest (RF), and Decision Tree (DT) algorithms applied to the same sample, which exhibited lower recognition rates of 69.32%, 74.88%, and 79.60%, respectively, highlighting the superiority of convolutional networks.

It is undeniable that the introduction of ICT resources leads to effective management of agricultural production in terms of efficiency, accessibility, and scalability. Reports by [19] demonstrate that the use of technology, despite initial resistance to innovation in the field, represents a low-cost framework for sanitary monitoring of cattle, based on behavioral activity images and muzzle patterns. The technique involves Support Vector Machines and Decision Tree classification for incremental decision-making and acceleration, aiming to mitigate noise, enhance contrasts, and improve quality. In other words, it is evident that to extract image details, a recognition system using machine learning and computer vision achieves higher accuracy for classification and identification, utilizing images sourced from servers containing encrypted muzzle databases, field databases, and health sensor databases.

The accuracy for recognizing cow movement behavior captured in natural environments using automated devices, smartphones, and other mobile equipment reached 95.00% when employing the VGG16 network for feature extraction and the Bi-LSTM (Bidirectional Long Short-Term Memory) classification model to reduce environmental interference. When applying four algorithms (C3D, VGG16-LSTM, ResNet50-LSTM, and DenseNet169-LSTM), the average accuracy for movement behavior recognition was 97.87%. To mitigate the trade-off between recognition rate and computational complexity, the use of an E3D (Efficient 3D CNN) algorithm was proposed. Compared with the enhanced ResNeXt network, ACTION-Net, and C3D-ConvLSTM, the latter was considered the most suitable for deployment on mobile devices [31].

Systems designed for mobile devices should be implemented with tools to minimize the trade-off between recognition accuracy and computational complexity due to the use of more efficient algorithms and neural network architectures. Employing quantization techniques to reduce the numerical precision of model weights or pruning to remove less important connections or neurons makes the model more compact and faster, while incurring minimal loss of accuracy.

Motivated by the goal of optimizing technology transfer for biometric muzzle recognition using surveillance cameras and wireless network technology, Kumar and Singh [11] proposed a method to maximize inter-class dispersion of muzzle point images while efficiently minimizing intra-class dispersion through the application of a muzzle point image scatter matrix. The effectiveness of the proposed recognition method reached 96.87% for individual cattle identification, with a processing time of 10.25 s for registration and recognition of the muzzle point, thereby creating a biometric database for cattle from diverse sources.

The bovine muzzle exhibits unique tissue characteristics, being a moist, thick, hairless surface with grooves and elevations specific to each animal, analogous to human fingerprints [19]. This structure, known as vibrissae, consists of keratinized hair extensions that function as sensory organs in some animals. Common among mammals and arranged around the muzzle, the complex or sinus hair follicle structure enables animals to assess the size of an object or determine the texture of a surface. Different species display variations in the shape and structure of vibrissae, although some properties are shared, such as the ability to function as a multicomponent force/torque sensor [34].

The presence of prominences on bovine muzzles has been documented in studies involving image capture and machine learning, with results validated through comparison and qualitative and quantitative analysis techniques [19]. However, despite advancements and future prospects highlighted by experimental and scientific studies, the widespread application of muzzle-based recognition techniques remains limited due to the relationship between spatiotemporal evolution and sensitive measurement caused by density.

To address this limitation, Abdellaoui et al. [35] suggests implementing a quantitative flow visualization method based on Background-Oriented Schlieren (BOS). This technique calculates gradients of different densities by analyzing the apparent displacement of the background pattern for each structure present in the images; thus, BOS was used to validate the digital identity of each studied bovine.

Although the use of chips or microchips for cattle detection and tracking is a more common practice, it depends on IoT-enabled wireless systems and is costly for large-scale animal identification and tracking. Similarly, the implementation of frameworks also requires hardware integration with the recognition system, trained personnel, and semi-automatic recognition systems based on muzzle images. Manual capture results in low similarity matching, achieving only 60% accuracy, and despite being a time-consuming process with image database processing, it is possible to perform fusion methods, such as combining facial and muzzle characteristics. For extracting multi-characteristics of the face and muzzle in images, techniques such as SURF, SIFT, Dense-SIFT+(x2), Eigen-face with PCA, and LDA were used, providing improved identification accuracy by delivering data related to the shape and structure of the muzzle and face image patterns, contour features, and texture information. The authors further state that the availability of these resources and the fusion of multi-characteristics achieve 98.56% accuracy in identification [19].

While the use of tags or microchips for cattle detection and tracking is effective, these systems rely on IoT-enabled wireless networks and become economically unfeasible for large-scale identification and monitoring. Furthermore, their implementation requires the integration of dedicated hardware with recognition systems, as well as trained personnel to operate semi-automatic models based on muzzle images. Manual image capture, in turn,

yields low similarity rates, typically achieving only about 60% accuracy, while also being a time-consuming process.

A challenge for integrating ICT with livestock management in cattle recognition and in using various methods to identify primary muzzle features is the availability of access to Bluetooth, WiFi, and RFID tools for more modern techniques. For traditional methods, access is often limited to online resources on some farms. Advanced technological resources, in terms of field accuracy and performance, require approaches for feature extraction, animal classification, and robust algorithms to achieve optimal results. The classification algorithms cited include logistic regression, Naive Bayes, Decision Trees, K-Nearest Neighbors, Support Vector Machine algorithms, random forests, and kernel methods. Superior results were achieved by applying SIFT, SURF, ORB (FAST-oriented and rotated BRIEF), and XGBoost algorithms [36].

Continuing the studies by [12], the CORF3D contour map test was introduced to suppress environmental noise, resulting in an accuracy of 99.64% for cattle recognition data. This represented a significant improvement over the 83.35% accuracy observed when using the Weber Local Descriptor (WLD) alone, and 96.5% after fusion with LBP and SVM. By combining a multi-level structure with the Inception ResNetV2 model, recognition accuracy increased to 98.21% for Holstein cattle. Additionally, the combination of FAST + SIFT + FLANN for body pattern recognition across 528 segments achieved an accuracy of 96.72%. Although multiple experimental setups produced accuracies above 90% with various algorithm combinations, only the use of the CORF3D contour map enabled results approaching 100%.

A study involving 4923 muzzle images collected from 268 beef cattle was evaluated using two classification models, ResNet50 and VGG16BN, with a maximum accuracy of 99.5% achieved using ResNet50. Approaches employing biometric identification systems with convolutional neural networks (CNNs) or Deep Belief Networks (DBNs) also demonstrated high accuracy, although they required large datasets and significant computational resources. In another study, the VGG16BN model was selected due to its superior baseline performance and its use of pre-trained ImageNet models. In this case, quantized and compressed data were generated to reduce data volume and redundancy. However, the inclusion of the discrete cosine transform (DCT) in the ResNet50 model resulted in an accuracy of 99.4%, surpassing that of the VGG16BN model [16]. The use of multiple models operating in tandem for identification tasks is a common practice to address scalability challenges associated with recognition methodologies, though the choice of model is always guided by the algorithm that provides the best results while maintaining flexibility to operate alongside other systems to minimize noise and errors.

Using a similarity scoring measure through Decision Tree classification, the matching and classification of structural features and the presence of vibrissae on cattle muzzles were performed, demonstrating an efficiency of 97.99% for the unique identification of each animal [30].

Techniques involving muzzle points have been employed to identify cattle with high recognition accuracy using deep learning. In parallel, DNA sequencing was conducted to verify pedigree and profiles in breeding animals, specifically cattle. While DNA analysis can indicate an individual's profile through the presence of SNPs (single nucleotide polymorphisms), this method is complex, costly, and time-consuming. In contrast, convolutional neural networks provide statistical validation for identification applications through deep learning, operating in real time, even though challenges may arise due to occlusion, nonlinear external image variations, or color similarity between animals [25].

Among the machine learning models aimed at cattle identification, commonly applied methods include SVM (Support Vector Machine), KNN (K-Nearest Neighbor), ANN (Artificial Neural Network), CNN (Convolutional Neural Network), ResNet (Residual Network), Inception, YOLO, and Faster R-CNN to characterize cattle coat patterns. These are complemented by LBP (Local Binary Pattern) and robust feature extraction techniques such as SURF, scale-invariant SIFT, Inception or CNN-based feature extraction, ResNet and Inception, and YOLO for detection. When used in combination, ResNet and VGG stand out by integrating three different detection models: YOLO, Faster R-CNN, and Mask R-CNN [10].

Experimental results based on a two-branch Convolutional Neural Network (TB-CNN) with images captured from different angles showed that the features from the two branches were fused through a global average pooling layer, combined with individual cattle classification. This fusion of feature extraction networks aims to enhance the extraction capability, while the pooling layer improves classification by reducing the number of network parameters [22].

The authors report that the recognition rates of the TB-CNN model reached 99.85% for Simmental cattle, 99.81% for Holstein cow faces, and 99.71% for a mixed dataset of cattle and cows. The parameterization allowed a comparison of the TB-CNN with other conventional networks, such as ResNet, which has well-recognized recognition performance, and the less common Beef-Data, Cow-Data, and Mix-Data networks. The integration of TB-CNN with these less common networks produced recognition rates of 99.70%, 98.07%, and 97.96%, respectively, with corresponding feature extraction rates of 99.70%, 98.07%, and 97.96%. When the TB-CNN feature extraction network was combined with the AlexNet model and OB-CNN (global average pooling layer), it enhanced the interaction between feature maps and categories, thereby improving recognition rates.

Overall, the experiments demonstrated high identification and extraction rates, all above 97%, and the combinations increased the robustness of the technique while reducing image capture noise for secure recognition. Notably, the three-part structure facilitated compression, excitation, and reweighting, allowing for feature recalibration and the learning of global information, while enabling interdependence among channels. By emphasizing the most important features of each channel, less significant features were suppressed, resulting in more efficient recognition outcomes.

Biometric systems based on visual muzzle recognition have been applied due to their ability to identify individual animals without the need for invasive techniques, while also providing improved protection and management across different species. This approach enables the classification of animals and their various categories, facilitating the study of populations within their natural habitats.

The primary biometric characteristics of cattle muzzle patterns enable analysis, tracking, and record-keeping for verification and parentage assignment for each animal, particularly those intended to meet reproduction, production, and distribution criteria in

beef cattle. The system captures images using field sensors and creates a database that stores the images through paired similarity techniques. Commonly, tools such as SIFT, SURF, and LBP are employed to extract information from the muzzle, and through PCA (Principal Component Analysis), LDA (Linear Discriminant Analysis), and ICA (Independent Component Analysis), it is possible to generate a digital fingerprint for species and individual identification. Algorithmic approaches further enable robotic systems, such as drones, to actively locate animals in various habitat scenarios, ensuring data quality and quantification [11].

The application of biometric image scanning of muzzle patterns was investigated by [21] using a three-stage system, which included object detection via YOLO v7, face and muzzle point detection, and finally data extraction and storage. The system achieved an average accuracy of 100% for face detection and 99.43% for muzzle point detection. Identification of the defined area patterns was performed using ORB and SIFT, either separately or in combination; subsequently, pattern matchers such as BFMatcher and FLANNMatcher were employed to identify each animal. A notable advantage of this system was the detection of faces and muzzle points in cows and buffaloes, achieving a mean Average Precision (mAP) of 99%.

Following the implementation of various structured systems after the positive results obtained with YOLOv7, a progressive decline in accuracy was observed. When using the Fuzzy-KNN algorithm, the system achieved an accuracy rate of 96.74%, corresponding to a 3.26% loss. With the BoHoG and LBP Histogram algorithms, accuracy decreased to 95.13%, reflecting a 4.87% loss. When VGGNet-DCNN and Cattle Face were applied as recognition methods, the accuracy further dropped to 70%. Returning to the YOLO-based system, the structured approach yielded a detection accuracy of 96.00%, distributed across 94% for training, 5% for validation, and 1% for testing.

Previously, Kusakunniran et al. [37] reported the use of a Haar cascade-based classifier for the identification of cattle using biometric images of the muzzle. The method began with locating the muzzle region in each image, followed by SIFT-based keypoint extraction from the patterns. Subsequently, the clusters were grouped according to these patterns using BoM and identified through an SVM (Support Vector Machine) model. When compared with public datasets of cattle slaughterhouse muzzle images and swamp buffaloes, the proposed method achieved 100% accuracy for cattle and over 90% accuracy for swamp buffaloes. The slightly lower accuracy for swamp buffaloes was likely due to animal movement and the presence of environmental artifacts, which introduced noise and reduced image clarity.

Multiple reports on practical precision monitoring in cattle using biometric methods indicate that the average detection accuracy of muzzle patterns was 99.13%, while identification accuracy reached 99.11% in tests. The two-stage YOLOv3-ResNet system demonstrated applicability for facial recognition across herds of any size, handling occlusions, multiscale scenarios, variable lighting, and differing image resolutions. To enhance the robustness and scalability of these systems, many models employ diverse approaches, including deep learning, Support Vector Machines, sparse group representations, fuzzy K-Nearest Neighbors, Fisher locality projections, Decision Trees, AdaBoost classifiers, SIFT matching, Bag-of-Visual Words, and classifier fusion. Additionally, leveraging transfer learning during data collection and model training has been shown to reduce processing time and improve overall system efficiency [6].

The lifecycle tracking of cattle, buffaloes, and cows, from birth to slaughter, can be achieved through identification using invariant features extracted from muzzle print images. As a validation measure, systems employing a Bag-of-600 Visual Words (BoVW) approach were tested, both standalone and in combination with two feature detectors: SURF and MSER (Maximally Stable Extremal Regions). Identification accuracy for SURF reached 75%, 83%, 91%, and 93% across the respective training data phases, whereas MSER yielded lower accuracies of 52%, 60%, and 67% for the same training phases [23].

In continuation of biometric studies, Ahmad et al. [21] utilized the Hadoop Distributed File System (HDFS) processed in parallel with Apache Spark clusters and the Python library PyTorch. The chosen network was YOLO, employing pre-trained weights from available datasets and bounding boxes for deep transfer learning. This approach involves training a new dataset by leveraging knowledge acquired from similar datasets. When an identifier is required and no reference database is available, FLANN is recommended for approximate matching of the animal's muzzle features with those in the provided database.

The Hanwoo cattle breed (Bos taurus coreana) is native to Korea and represents the most widely consumed source of beef in the region. They are typically raised on small pastures due to limited land availability for breeding purposes. Using a YOLOv8-based model, 150 randomly selected muzzle images were stored for validation testing. Transfer learning was performed with EfficientNet v2, employing the SGD, RMSProp, Adam, and Lion optimizers. The Lion optimizer achieved the highest validation accuracy, reaching 0.981 after 36 epochs and 12 transfer learning models, while Adam reached a test accuracy of 0.970, and RMSProp exhibited the lowest repeated error. A limitation of this study was the use of smartphone cameras for image capture, which increased variability in image quality and consistency, being particularly influenced by changes in angles and environmental factors [24].

When conducting real-time experiments with animals, variations in illumination, nonlinear movement, and occlusion present significant challenges. The recommended approach for assessing cows' body patterns involves an input system that processes an image and outputs a classification into predefined categories. For training, a standard dataset of 1000 images encompassing 10 cow breeds was processed through a series of convolutional layers with filters, pooling layers, fully connected layers, and a softmax function for image classification. The training and testing achieved identification and accuracy rates of 92.59% and 89.5%, respectively. The convolutional layers primarily preserved pixel relationships, learning according to the input data. For very large images, pooling layers were applied to reduce the number of parameters and dimensionality, employing techniques such as Deep Belief Networks (DBNs), Stacked Denoising Autoencoders (SDAEs), Contrast Limited Adaptive Histogram Equalization (CLAHE), Gaussian filtering, and Local Binary Patterns (LBPs) [38].

A total of 59 images were comparatively classified for individual cattle identification, and the weighted cross-entropy loss function was found to enhance the accuracy of individual identification. Test accuracies and processing speeds ranged from 1.2% to 98.4% using ShuffleNetV2, with the lowest performance observed for VGG16BN. To ensure model reliability, a Pearson correlation

coefficient matrix (R) was introduced to better understand the relationships among model parameters [5].

The results confirmed the hypothesis that a more complex model with a higher number of parameters tends to achieve greater accuracy; however, longer processing times were also noted. Additionally, animals with more than four muzzle images reached over 90% identification accuracy, suggesting that a quantitative limit of five images per animal would provide an appropriate reference parameter.

Continuous monitoring of the diverse needs of each animal remains a significant challenge in precision livestock farming, which has increased interest in adopting 3D techniques, including 3D sensors and more complex data processing, although 2D tools are still generally recommended for automated assessment. When analyzing the challenges that persist for the future development of livestock technologies, it is evident that these techniques can similarly be implemented in the management of other farmed animal species [39].

Techniques such as Hot Branding, Freeze Branding, Paint Branding, and RFID systems have been applied for the identification of Kamphaeng Saen cattle for registration and traceability, using muzzle impressions as a biometric approach. Two algorithms were employed to compose the system: SIFT for detecting key points and random samples, and RANSAC to remove outliers and enhance the robustness of image matching. Cattle identification was determined based on a similarity score, and the superiority of the method was achieved through the coupling of SIFT with RANSAC, providing robustness against outliers and yielding an identification accuracy of 92.25% [26].

Applying three CNN architectures for the identification of Pantaneiro cattle, de Lima Weber et al. [17] conducted experiments using DenseNet-201, ResNet50, and Inception-ResNet-V. All networks were subjected to 10-fold stratified cross-validation over 50 epochs. The results demonstrated an average accuracy of 99%, regardless of the animals' angles or positions. Therefore, it can be stated that all classes achieved a high number of correct identifications, and none of the proposed models showed statistically significant differences among them. ANOVA hypothesis testing in RStudio with p-values < 0.05 was used for each metric, indicating that the models are reliable for cattle recognition.

Subsequently, using the Shi-Tomasi detector for cattle identification and the SIFT and SURF feature descriptors to extract features from digital images in linear dimensionality, three classification techniques were applied—MLP (Multilayer Perceptron), Random Forest, and Decision Tree—on a dataset comprising 930 muzzle image patterns from four cattle breeds. The combination of Shi-Tomasi with SURF and SIFT, implemented to achieve recognition accuracy, yielded results of 69.32%, 74.88%, and 79.60%, respectively. The highest accuracy of 83.35% was reported when the Bootstrapping aggregation methodology (Ag) and the Random Forest classifier were employed in combination with Shi-Tomasi, SURF, and SIFT [36].

Proposing an evaluation of a cattle identification system from a top-down perspective (dorsal view), Sharma et al. [27] utilized a dataset and pre-trained weights for immediate reproducibility with images captured by a 3D camera. Based on CNN and MLP architectures, which do not require species-specific coat patterns or close-up muzzle impressions for operation, the images were clustered using KNN for highly accurate identification, eliminating the need to retrain the network to register new images. Two backbone architectures were also assessed, ResNet and PointNet, specialized for operating on 3D point clouds, alongside CowDepth2023, to evaluate color-depth images and dorsal spines, including fully black and fully white breeds.

The ResNet698 color backbone achieved a k-NN identification accuracy of 99.97%, while PointNet reached 99.36% using only dorsal depth maps. Despite substantial interclass variability in body shape, Grad-CAM701 (Gradient-weighted Class Activation Mapping) and PC-SM (Point Cloud Saliency Mapping) enabled individual cows to be identified from RGB images, producing DNNs and MLPs capable of recognizing cattle through depth images.

The studies indicate that, with respect to accuracy, the reported results consistently exceed an average of 95%, regardless of the adopted architecture, as summarized in **Table 3**. This convergence suggests that performance is predominantly influenced by the employed techniques and their suitability for the target application rather than by the specific datasets used. Differences among the studies are therefore more meaningfully explained by variations in methodological choices, processing efficiency, and practical constraints, such as robustness to noise and illumination variability, acquisition angles, and the scale of the image sets, which directly reflect the intended application, including facial and muzzle identification, body and breed pattern recognition, and bovine facial analysis.

Table 3 • Comparison of algorithmic systems applied to cattle muzzle recognition with reported accuracy above 95%.

Method	Precision	Processing speed	Scenario	Quantity of animals (average)
YOLO (You Only Look Once)	99.43–99.7%	High Velocity	Large herds with variable lighting conditions and multi-scale features	268
SVM (Support Vector Machine)	96.74–100%	Moderate	Identification of facial and muzzle features, combined with SIFT	Varied
Convolutional Neural Network (CNN)	99.5–99.85%	Moderate	Classification of muzzle and facial features using images captured from different angles	243
Faster R-CNN	95–98.21%	Moderate and Slow	Identification of body and facial patterns in controlled environments	Varied
ResNet50	99.5%	Moderate and Slow	Facial and muzzle feature extraction with large-scale datasets	4923
KNN + SIFT + SURF	96.5–99.5%	Moderate	Identification of specific muzzle patterns with robustness	Varied

4. Conclusions

This review systematically assessed neural network-based architectures for cattle identification and recognition, with emphasis on facial and muzzle regions due to their distinctive and stable anatomical characteristics. The surveyed literature demonstrates that both classical feature-based methods, such as SIFT, SURF, and KNN, and deep learning architectures from the YOLO and ResNet families are capable of extracting discriminative representations for individual identification, with convolutional neural networks consistently forming the core of high-performing systems and achieving recognition accuracies above 90%, and near-optimal results in controlled scenarios, as reported for recent YOLOv8 implementations.

Importantly, a critical synthesis of the reviewed studies reveals that robust bovine facial and muzzle recognition depends primarily on a consistent set of visual characteristics, including fine-grained texture patterns of the muzzle surface, the spatial configuration of nostrils, local curvature cues, and hierarchical feature representations learned by deep convolutional layers. These characteristics emerge recurrently across different architectures and application contexts, indicating that feature relevance, rather than model selection alone, is central to system performance. This insight directly addresses the need for more substantive findings by moving beyond accuracy reporting toward an understanding of which visual cues underpin successful recognition.

Building on this analysis, future research should explicitly investigate the contribution of individual visual features within architectures such as YOLOv8 and ResNet-based models, either through targeted ablation studies or structured feature attribution analyses, using available datasets. In parallel, the development of computationally efficient and interpretable models, coupled with more standardized evaluation protocols, is essential to facilitate practical adoption. Addressing these directions will bridge the gap between methodological advances and real-world deployment, enabling the effective integration of intelligent vision-based systems into routine livestock management and supporting scalable innovation in precision agriculture.

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Author contributions

Conceptualization, K.R.d.A.P. and J.V.d.A.P.; methodology, K.R.d.A.P. and J.V.d.A.P.; validation, V.A.d.M.W. and F.d.L.W.; formal analysis, K.R.d.A.P., F.d.L.W. and S.L.G.; investigation, K.R.d.A.P. and J.V.d.A.P.; resources, V.A.d.M.W. and F.d.L.W.; data curation, K.R.d.A.P.; writing—original draft preparation, K.R.d.A.P. and J.V.d.A.P.; writing—review and editing, K.R.d.A.P., J.V.d.A.P. and S.L.G.; visualization, J.V.d.A.P. and F.d.L.W.; supervision, V.A.d.M.W. and F.d.L.W.; project administration, V.A.d.M.W. and F.d.L.W.; funding acquisition, V.A.d.M.W. and F.d.L.W. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

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Data availability statement

All data supporting the findings of this publication are available within this article and/or its supplementary materials.

Supplementary materials

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